

Plant disease recognition when a robot takes the picture

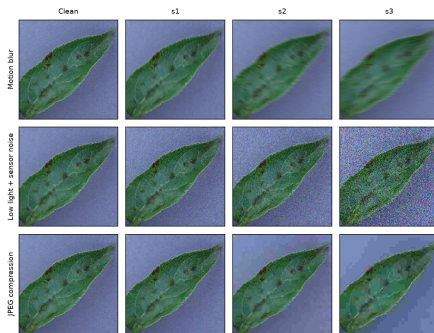
Task. 38 plant disease classes, 14 crops, PlantVillage — laboratory capture: one detached leaf, uniform background, even light. 41 275 / 10 861 / 2 169 images; the test split is carved by *moving* 5 % of each training class, so no leakage.

Two transfer strategies, one controlled variable — which parameters get gradients:

- **Frozen backbone** — linear probe, 77 862 trainable parameters
- **Fine-tuned** — all 23 585 894 trainable

Question. A pretrained ResNet-50 saturates on clean PlantVillage, so the question is not whether transfer works but **which strategy survives leaving the laboratory**.

We model a field phenotyping robot: motion blur while driving, low light with sensor noise under the canopy, JPEG compression for radio transmission — three severities each, applied to the test set only, no retraining.



Rows: motion blur, low light + noise, JPEG.

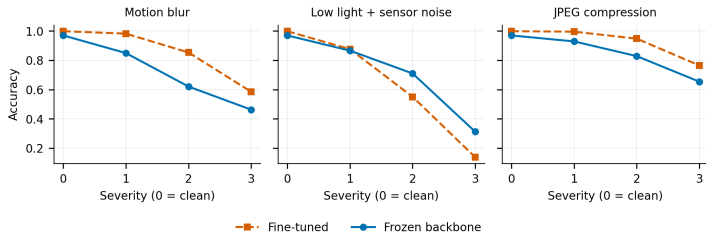
Columns: clean, then severity 1 to 3.

Clean data says one thing, degraded data says another

Clean test	Froz.	F.-t.
Accuracy	.970	.999
Macro F1	.966	.998
Errors	66	3
Train. params	78 k	23.6 M
Epochs	47	16
Time (min)	14.7	9.9

Same graph, same parameter count $\Rightarrow$ identical inference cost: 4.74 ms/image at batch 1, 0.31 ms at batch 64, 773 MB on an A100.

LR ablation $10^{-4}/10^{-3}/10^{-2}$:
.9986 / .9991 / .9963 — the comparison is not an artefact of the learning rate.



Fine-tuning wins under blur and JPEG at every severity, by up to 23 points. Under **low light with sensor noise** the **curves cross**: the frozen backbone is 16 points ahead at severity 2 and 17 at severity 3.

Fine-tuning sharpens early filters onto the high-frequency statistics of noise-free laboratory capture. Blur and JPEG *remove* high frequencies; sensor noise *injects* random energy into exactly that band.

What breaks, and what to take away

Every clean error is a within-crop confusion — all 3 of the fine-tuned model and all 66 of the frozen one. Cross-crop error rate: exactly zero. Corn Cercospora → Northern Leaf Blight, Tomato Bacterial spot → Late blight, Target Spot ↔ Spider mites. The fine-tuned model's corn errors are leaves so far into necrosis that the distinguishing lesion geometry is gone.

Wrong, and confident about it. Mean confidence on errors: 0.86 fine-tuned against 0.53 frozen. Under motion blur severity 3 it still reports 0.86 while being right 58.5 % of the time — it does not signal that it has left its operating regime.

Fine-grained classes collapse first. At low light severity 2 the fine-tuned model drops Potato healthy, Cedar apple rust and Tomato Bacterial spot from $F1 \approx 1.0$ to 0. “No lesion” becomes indistinguishable from “lesion hidden by noise”.

Conclusion.

- Fine-tuning is the right default — 3 errors against 66, in a third of the epochs.
- But robustness is **artifact-dependent**: it is not a single number, and the ordering reverses under sensor noise.
- Pick the transfer strategy for the capture regime you expect, and report calibration, not only accuracy.

Limitations. Corruptions are simulated; focus, perspective and background are not modelled. PlantVillage repeats the same physical leaf, so clean accuracy overstates generalisation. One architecture, one dataset.

[github.com/EgorSavchenko-web/
icv2026-plant-disease](https://github.com/EgorSavchenko-web/icv2026-plant-disease)